

CENTRAL NEBRASKA, NE, USA

WMO: 725520

Lat: 40.961N

Lon: 98.314W

Elev: 561

StdP: 94.77

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14935

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.4	-16.7	-23.6	0.5	-18.7	-21.0	0.6	-16.0	15.2	-4.5	13.7	-3.3	4.5	350	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	35.3	23.5	33.6	23.0	31.9	22.4	25.4	31.8	24.6	31.1	23.6	30.0	6.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	19.7	29.2	22.6	18.5	28.3	21.7	17.5	27.4	81.2	31.9	77.4	31.2	73.6	30.0	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.0	11.6	10.4	-24.2	38.6	2.1	1.9	-25.7	40.0	-27.0	41.1	-28.2	42.1	-29.7	43.5		
(5)			-24.5	27.3	2.1	0.8	-26.0	27.9	-27.2	28.4	-28.4	28.8	-29.9	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-3.7	-1.8	4.4	10.2	16.3	22.3	24.8	23.3	19.0	11.6	4.2	-2.2		
	DBStd	11.37	6.76	6.99	6.66	5.50	4.92	3.90	3.34	3.17	4.70	5.29	5.78	6.18		
	HDD10.0	1629	424	331	194	63	7	0	0	0	2	44	188	377		
	HDD18.3	3362	682	563	431	248	99	11	1	3	47	217	424	635		
	CDD10.0	1913	0	2	22	69	202	368	457	414	273	93	13	0		
	CDD18.3	604	0	0	1	5	36	129	200	158	68	7	0	0		
	CDH23.3	6293	0	1	18	91	422	1367	2107	1455	727	101	4	0		
	CDH26.7	2643	0	0	2	25	148	575	981	605	278	28	1	0		
(14) Wind	WSAvg	5.0	5.0	5.3	5.6	6.0	5.3	4.8	4.0	3.9	4.6	5.0	5.0	4.9		
(15) Precipitation	PrecAvg	653	14	20	40	67	102	101	78	72	66	44	26	20		
	PrecMax	992	42	86	168	186	230	355	271	222	229	152	96	88		
	PrecMin	293	0	2	0	2	11	1	4	15	5	0	1	1		
	PrecStd	157	10	16	36	37	56	69	52	44	51	35	24	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.7	20.3	25.7	29.9	33.5	36.3	38.0	36.8	34.7	30.3	23.5	15.6		
		MCWB	7.7	10.1	13.4	16.7	20.0	22.6	23.9	23.1	21.1	18.3	12.8	7.8		
	2%	DB	11.3	14.8	22.0	25.9	30.4	33.9	35.7	34.1	32.2	25.9	19.6	11.8		
		MCWB	5.2	7.6	12.0	15.4	19.4	22.6	24.4	23.6	20.9	15.9	10.8	6.3		
	5%	DB	8.1	11.3	18.3	22.9	28.0	32.0	33.7	32.0	30.0	23.3	16.2	9.1		
		MCWB	3.6	6.1	11.0	13.8	18.2	22.0	23.9	23.4	20.3	14.7	9.7	4.5		
	10%	DB	5.2	8.0	15.1	19.9	25.3	30.0	31.8	30.1	27.6	20.7	13.1	6.2		
		MCWB	2.1	4.2	9.2	12.5	17.2	21.1	23.4	22.5	19.0	13.7	8.0	2.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.9	11.2	16.5	18.5	22.4	25.3	26.9	26.0	23.6	20.1	14.9	9.8		
		MCDB	15.5	17.5	20.8	25.9	29.5	31.7	33.1	32.0	30.6	26.4	19.0	14.3		
	2%	WB	5.6	8.3	14.1	16.5	20.9	24.1	25.8	25.0	22.4	18.1	12.3	6.5		
		MCDB	10.6	14.2	19.2	23.0	27.6	31.3	32.3	31.2	29.1	22.8	17.7	11.2		
	5%	WB	3.9	6.2	11.7	14.9	19.6	23.1	25.0	24.1	21.5	16.2	10.2	4.7		
		MCDB	7.7	11.2	17.5	21.7	26.0	29.9	31.6	30.2	27.8	21.3	15.7	8.7		
	10%	WB	2.3	4.2	9.4	13.2	18.1	22.1	24.1	23.1	20.4	14.6	8.1	2.8		
		MCDB	5.1	7.9	15.1	19.1	23.9	28.5	30.6	29.0	25.9	19.3	13.0	6.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	11.2	12.8	12.9	12.1	12.3	11.8	11.7	13.4	13.2	12.4	10.6		
		MCDBR	15.3	16.2	18.3	17.9	15.8	14.6	14.1	14.1	15.3	17.1	17.3	14.7		
		MCWBR	10.1	10.1	10.2	9.3	7.0	6.0	6.0	6.0	6.4	8.5	10.0	9.8		
	5% WB	MCDBR	14.6	15.7	16.1	15.8	13.9	12.9	12.6	12.4	13.2	13.6	16.0	14.1		
		MCWBR	9.9	10.2	9.5	9.2	7.0	6.2	6.4	6.1	6.3	7.8	9.7	9.9		
(40) Clear-Sky Solar Irradiance	taub		0.274	0.280	0.307	0.345	0.370	0.385	0.400	0.397	0.362	0.320	0.281	0.271		
	taud		2.483	2.494	2.452	2.364	2.368	2.375	2.349	2.348	2.404	2.504	2.524	2.495		
	Ebn at Noon		897	944	948	927	905	889	872	864	875	883	882	872		
	Edh at Noon		74	86	100	118	121	120	122	119	105	84	71	68		
(44) All-Sky Solar Radiation	RadAvg		2.19	3.04	4.08	5.16	5.99	6.85	6.74	5.84	4.75	3.40	2.42	1.85		
	RadStd		0.20	0.20	0.35	0.28	0.38	0.43	0.32	0.35	0.30	0.34	0.17	0.18		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page